

Dilations and Similar Figures in Practice

Answer Key

High School Geometry

Quick Answers

- | | |
|---|--|
| 1. x-coordinate of $A' = 3$, y-coordinate of $A' = -2$ | 6. 9.00 m |
| 2. 2.5 | 7. x-coordinate of $D' = 14$, y-coordinate of $D' = -5$ |
| 3. 340 cm | 8. (a) perimeter of the scaled courtyard = 100 m, (b) area of the scaled courtyard = 600 m^2 , — |
| 4. (a) x-coordinate of $P' = 6$, (a) y-coordinate of $P' = 3$, (b) $PQ = 5$, (c) $P'Q' = 15$ | |
| 5. 25 | |
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What is verified

14 of 15 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problem 8. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-SRT.A.1 – *problems 1, 4, 7*

HSG-SRT.A.2 – *problems 2, 5*

HSG-SRT.B.5 – *problems 3, 6, 8*

Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

2. *If they got 18: subtracted the two widths instead of dividing*

5. *If they got 0.25: divided without converting the real length to centimetres first*

6. *If they got 20.25: set the proportion up the wrong way round, dividing by the student's height instead of the student's shadow*

1. $A(6, -4)$ dilated about the origin by factor 0.5. Find A' .

Solution: A dilation about the origin multiplies each coordinate by the scale factor.

$$x' = 0.5 \times 6 = 3, \quad y' = 0.5 \times (-4) = -2$$

So the coordinates are

and

-2

A factor between 0 and 1 shrinks the figure: A' is half as far from the origin as A , in the same direction.

- 2.** Photograph 12 cm wide enlarged to 30 cm wide. Find the scale factor.

Solution: A scale factor is a ratio of matching lengths, image over original — a division, never a difference.

$$k = \frac{30}{12}$$

So

$k = 2.5$

- 3.** Blueprint at 1 cm to 40 cm; a wall measures 8.5 cm on the plan. Find the real length.

Solution: Each drawn centimetre stands for 40 real centimetres, so multiply the drawn length by 40.

$$8.5 \times 40$$

So

340 cm

- 4.** $P(2, 1)$, $Q(6, 4)$ dilated about the origin by factor 3; $Q'(18, 12)$.

Solution (a): $x' = 3 \times 2 = 6$ and $y' = 3 \times 1 = 3$, so the coordinates are

6

and

3

Solution (b): $PQ = \sqrt{(6-2)^2 + (4-1)^2} = \sqrt{16+9} = \sqrt{25}$, so

$PQ = 5$

Solution (c): $P'Q' = \sqrt{(18-6)^2 + (12-3)^2} = \sqrt{144+81} = \sqrt{225}$, so

$P'Q' = 15$

The image length is exactly 3 times the original — which is what "scale factor 3" means, and it is the one property that separates a dilation from a rigid motion.

- 5.** Model car 18 cm long; real car 450 cm long. How many times longer is the real car?

Solution: Both lengths must be in the same unit before dividing, and they are — centimetres.

$$\frac{450}{18}$$

So the real car is
times longer.

25

- 6.** Student 1.6 m tall with a 2.4 m shadow; flagpole shadow 13.5 m. Find the flagpole's height.

Solution: The two right triangles are similar, so matching sides are in the same ratio: height over shadow is the same for both.

$$\begin{aligned} \frac{h}{13.5} &= \frac{1.6}{2.4} \\ h &= \frac{1.6 \times 13.5}{2.4} = \frac{21.6}{2.4} \end{aligned}$$

So

$$h = 9.00 \text{ m}$$

Sense check: the flagpole's shadow is about 5.6 times the student's, and 9 m is about 5.6 times 1.6 m ✓

7. $D(5, 1)$ dilated about $C(2, 3)$ by factor 4. Find D' .

Solution: With a centre other than the origin, measure the offset from the centre, scale that offset, and add it back to the centre.

$$x' = 2 + 4(5 - 2) = 2 + 12 = 14$$

$$y' = 3 + 4(1 - 3) = 3 - 8 = -5$$

So the coordinates are

$$14$$

and

$$-5$$

Multiplying the raw coordinates instead would give $(20, 4)$ — the right size but the wrong place, because that dilates about the origin.

8. Courtyard 12 m by 8 m scaled by 2.5. (a) Perimeter. (b) Area. (c) Why they differ.

Solution (a): The original perimeter is $2(12 + 8) = 40$ m, and a perimeter is a sum of lengths, so it is multiplied by 2.5 once.

$$40 \times 2.5$$

So

$$100 \text{ m}$$

Solution (b): The original area is $12 \times 8 = 96 \text{ m}^2$. Both the length and the width are stretched by 2.5, so the area picks up the factor twice, $2.5^2 = 6.25$.

$$96 \times 6.25$$

So

$$600 \text{ m}^2$$

Solution (c) — instructor-judged. A model response: a perimeter is a sum of lengths, and every length is multiplied by 2.5, so the sum is too. An area is a product of two lengths, and both of them are multiplied by 2.5, so the area is multiplied by $2.5 \times 2.5 = 6.25$. One scale factor, two different multipliers, because of how many lengths each measurement is built from.

Full credit: distinguishes a sum of lengths from a product of lengths and gives both multipliers. *Half credit:* gives 2.5 and 6.25 with no reason. *No credit:* says only that area grows faster.